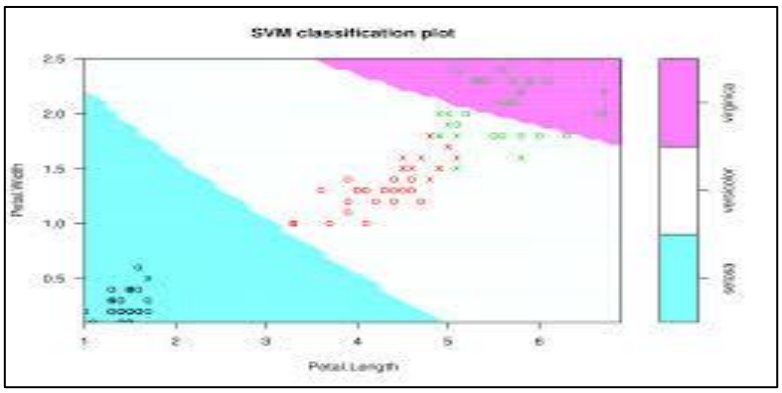
Iris Versicolor **Iris Setosa** **Iris Virginica**

Implementation of Classification Algorithms using Iris Dataset

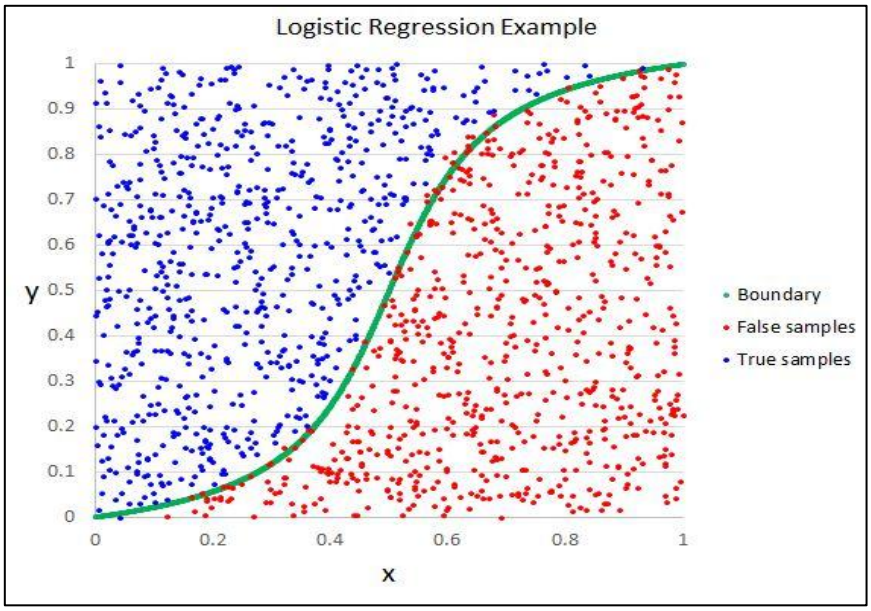
- We have used Iris Dataset which have 6 columns with sepal length, sepal width, petal length and petal width, Id and Species columns with 150 rows.
- The dataset is well structured and prepared to directly work on Algorithms. So in this dataset, ***['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width']*** are the input features which are also called as independent columns which helps us to classify the iris type. And the output column will be ***['Species']*** which contains 3 categories namely ***iris-setosa, iris-virgina, iris-versicolor.***
- We have performed feature transformation using StandardScaler from sklearn. It will normalizes the whole dataset so that it omits the biased or unbiased results.

1. Support Vector Machine (SVM) :



A Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that best separates data points of different classes in a feature space. The key idea is to maximize the margin — the distance between the hyperplane and the nearest data points from each class, called support vectors. When data is not linearly separable, SVM uses the kernel trick to map input features into a higher-dimensional space where a linear boundary can be found. Common kernels include linear, polynomial, RBF (Radial Basis Function), and sigmoid. SVM is effective in high-dimensional spaces, making it well-suited for text classification and image recognition tasks. SVMs are robust to overfitting, especially in cases where the number of features exceeds the number of samples.

2. Logistic Regression:



Logistic Regression is a supervised machine learning algorithm used for classification tasks, mainly binary classification. It predicts the probability of an input belonging to a class using the sigmoid function, which outputs values between 0 and 1. If the probability is above 0.5, the input is assigned to one class, otherwise to the other. It models the relationship between input features and the target variable using a linear equation. The model is trained by minimizing the log-loss using optimization methods like gradient descent. It can also handle multi-class classification using techniques like One-vs-Rest or Softmax. Regularization methods like L1 and L2 help prevent overfitting. It is simple, interpretable, and computationally efficient, making it a popular baseline model in machine learning.

3. Gaussian Distribution:

Gaussian Distribution, also known as the Normal Distribution, is one of the most common and important probability distributions in statistics and machine learning. It describes how data values are spread around a central mean value in a bell-shaped curve. Most of the data points tend to cluster around the mean, and fewer points appear as you move further away from it. It is defined by two parameters — the mean (μ), which controls the center of the curve, and the standard deviation (σ), which controls the width or spread. A smaller standard deviation means the data is tightly packed around the mean, while a larger one means it is more spread out. The distribution follows the 68-95-99.7 rule, meaning about 68% of data falls within one standard deviation, 95% within two, and 99.7% within three. It is symmetric around the mean, meaning the left and right sides of the curve are mirror images. Gaussian distribution appears naturally in many real-world scenarios such as heights, weights, test scores, and measurement errors. In machine learning, it is used in algorithms like Naive Bayes, Gaussian Mixture Models, and Linear Discriminant Analysis. It is a foundational concept in statistics and forms the basis of many machine learning and deep learning techniques.

4. ***Decision Tree Classifier:***

A Decision Tree Classifier is a supervised machine learning algorithm that makes decisions by splitting data into branches based on feature conditions, similar to a flowchart. It starts from a root node, which represents the entire dataset, and splits it into smaller subsets based on the most important feature. Each internal node represents a condition or question on a feature, and each branch represents the outcome of that condition. The splitting continues until it reaches a leaf node, which gives the final predicted class label. The best feature to split on is chosen using measures like Gini Impurity or Information Gain (Entropy). Decision trees are very easy to understand and visualize, making them one of the most interpretable machine learning models. They can handle both numerical and categorical data without requiring much data preprocessing. However, decision trees tend to overfit the training data if they are allowed to grow too deep without any restrictions. To avoid overfitting, techniques like pruning, setting maximum depth, or minimum samples per leaf are used. Decision trees also serve as the building blocks for powerful ensemble methods like Random Forest and Gradient Boosting**.**

5. ***Random Forest Classifier:***

A Random Forest Classifier is a supervised machine learning algorithm that builds multiple decision trees and combines their results to make a final prediction. It is based on the concept of ensemble learning**,** where many weak models are combined to form a stronger and more accurate model. During training, it creates a large number of decision trees, each trained on a random subset of the data and features**.** This randomness helps in making each tree different from the others, reducing the chances of overfitting. For classification, each tree gives a predicted class, and the final output is decided by majority voting among all the trees. Random Forest is highly accurate and performs well even on large and complex datasets without much tuning. It is robust to **o**utliers and missing values**,** making it suitable for real-world messy data. It also provides a useful feature called feature importance, which tells us which features contribute the most to the prediction. However, it is less interpretable than a single decision tree because it involves hundreds of trees working together. Random Forest is widely used in applications like fraud detection, medical diagnosis, stock market prediction, and image classification**.**

Conclusion:

Finally, I conclude that in iris dataset we have three categories and our five models are well trained on the training set and has given highest accuracy as below:

- For Logistic Regression Model we got the accuracy as 100 percent, where model can predict accurately and its learning rate is high
- For Gaussian Distribution from Naïve Bayes, we got the accuracy as 100 percent, where model can predict accurately and its learning rate is high
- For Decision Tree Classifier, we got accuracy of 97 percent, where model can predict perfectly and its learning rate is slightly higher
- For Random Forest Classifier, we got accuracy of 100 percent, model can predict correctly and learning rate is higher. And SVM has 97 percentage of accuracy.